MTH101 Mcqs Final Term By Vu Topper RM
For More Help Contact 03224021365,03164980094

What does it mean for an improper integral to converge?

A. The integral has a well-defined upper and lower bound.
B. The function f(x) is always positive.
C. The limit of the integral approaches a finite value.
D. The limit of the integral approaches infinity.

If a flat surface of area A is submerged horizontally in fluid at a depth h then the fluid exerts a force F perpendicular to the surface this force F is defined as

A. F =A ρh
B. F = ρhA
C. F =ρha
D. F = ρh

If a function is constant over an interval, what can be said about its arc length on that interval?

A. The arc length is proportional to the slope.
B. The arc length is equal to the width of the interval.
C. The arc length is infinite.
D. The arc length is zero.

If [-8,8] is subdivided into ‘16’ equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be--------.

A. 2
B. 4
C. 3
D. 5

MTH101 Mcqs Final Term By Vu Topper RM
For More Help Contact 03224021365,03164980094

The value of 2f1 dx=----------

A. 1
B. 3
C. 0
D. 2

Arc length of the curve y=1 from x=a to x=b is .....

A. 2(b-a)
B. a-b
C. b-a
D. 0

Distance between (3,-2) and (4,0) using the distance formula is…….

A. None of these.
B. sqrt (5)
C. sqrt (3)
D. 0

An object moves 4 ft. along a line while subjected to a force of 200 Ibs in its direction of motion. The work done is

A. 600 ft. Ibs
B. 800 ft. Ibs
C. 200 ft. Ibs
D. 400 ft. Ibs

Derivative of f(x)= a-7,where a is a constant is....

A. 0
B. -7
C. a
D. a-7

MTH101 Mcqs Final Term By Vu Topper RM
For More Help Contact 03224021365,03164980094

Which of the following statements is true regarding the ARC LENGTH PROBLEM?

A. It only deals with straight lines.
B. It is applicable only to polynomial functions.
C. It focuses on finding the area under a curve.
D. It addresses the challenge of determining the length of curved lines.

The volume V of a cylinder with base area A and height h is calculated by ---------.

A. V = A h^2
B. V = A h
C. V = A h^3
D. V = 2A h

An object moves 6 ft. along a line while subjected to a force of 500 Ibs in its direction of motion. The work done is

A. 600 ft. Ibs
B. 3000 ft. Ibs
C. 300 ft. Ibs
D. 500 ft. Ibs

Distance between the points (-2,-3) and (-4,-1) using the distance formula is.....

A. Sqrt(10)
B. Sqrt(8)
C. 0
D. None of these

MTH101 Mcqs Final Term By Vu Topper RM
For More Help Contact 03224021365,03164980094

1.If Sin(3x^2) / 6 + C is the anti–derivative of a function f(x), then f(x) = _____.

x^2 Cos(3x^2).

x Cos(3x^2).

x Cos(3x).

none

2.Which of the following is the integral of Sin(3x+5) with respect to x ?

-1/3[-Cos(3x+5)]

- **1/3[-Cos(3x+5)]**

1/15[-Cos(3x+5)]

-Cos(3x+5)

3.If 'n' goes from 1 to 3 and the summation of 'na' = 6a, then the value of 'a' is ----------

- **6**

-6

1

undetermined

4.If 'n' goes from 1 to any large ODD number then the summation of '(-1)^n' = ---------

- **-1**

0

1

that specific large ODD number

5.1+2+3……+t equals

n(n+1)/2

- **t(t+1)/2**

n(n+1)(2n+1)/6

none

6.If definite integral of f(x)=Sinx over [a,0] is equal to '-2' then the value 'a' is---------

pi/2

- **pi**

0

-pi

7.If the definite integral of f(x)=3 over [1,x] is greater than '12' then -----

x>12

x>5

- **x>3**

x>1

8.If [-8,8] is subdivided into '16' equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be--------.

2

3

4

- **5**

9.Which of the following is the integral of sin(2x)?

cos(2x)+C

2cos(2x)+C

- **−(1/2)cos(2x)+C**

none

10.Sum of cubes of n-terms of a series whose nth term is 'n' = ---

Square of n(n+1)(2n+1)/6

Square of n(n+1)/2

- **Square of (n+1)/2 (just a guess)**

Square of n(n+1)/6

Arc length of the smooth curve x=g(y) for y=a to y=b is.....
Select correct option:
L=∫ab1+(dxdy)2−−−−−−−−√dy
L=∫ab1+(dydx)2−−−−−−−−√dx

Question # 2 of 10 (Start time: 02:48:15 PM) Total Marks: 1
The integral of the sum of two functions is equal to the sum of their integrals.
Select correct option:
True
False

Question # 3 of 10 (Start time: 02:48:51 PM) Total Marks: 1
Evaluate∫x0sintdt=
Select correct option:

1+cosx

1−cosx

1+cost

1+cost

Question # 5 of 10 (Start time: 02:51:41 PM) Total Marks: 1

Mathematically second fundamental theorem of calculus can be written as,

Select correct option:

ddx∫taf(t)dt=f/(x)

ddx∫taf(t)dt=f(x)

ddx∫taf(t)dt=f(t)

none of these

Question # 6 of 10 (Start time: 02:53:23 PM) Total Marks: 1

Ifthecurveover[a,b]isrevolvedabouty−axis,thenthevolumeiscalculatedbytheformula−−−−−−−

Select correct option:

∫abπ[f(x)]2dx

∫abπ[f(y)]2dy

Question # 7 of 10 (Start time: 02:54:16 PM) Total Marks: 1

Theareaoftheellipsex2a2+y2b2=1

Select correct option:

π(a+b)

πab

14π(a2+b2)

Noneofthese

MTH101 Mcqs Final Term By Vu Topper RM
For More Help Contact 03224021365,03164980094

Question # 8 of 10 (Start time: 02:56:00 PM) Total Marks: 1
Arc length of the curve y=1 from x=a to x=b is .....
Select correct option:
2(b-a)
b-a
a-b
0

Question # 9 of 10 (Start time: 02:57:18 PM) Total Marks: 1
The volume of cylindrical shell for R = 5, r = 3 and h = 2 is ----------
Select correct option:
25*pi
30*pi
32*pi
36*pi

To get better approximation to actual area under a continuous curve over a closed interval, we have to increase .....

Subintervals